

Nearly linear face counts for exact three- and four-wise fairness

A consequence of permutation designs and palindrome composition

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Let $M_k(n)$ be the least common number of faces of n uniform dice, with all labels distinct, for which every subset of at most k dice has uniform relative order. Here k is fixed while n grows.

Theorem 1. *For $n \geq 5$,*

$$\frac{n+3}{2} \leq M_3(n) \leq M_4(n) \leq Cn(\log n)^2$$

for an absolute constant C . More precisely,

$$M_3(n) \leq (6\sqrt{e} + o(1))n(\log_2 n)^2, \quad M_4(n) \leq (12\sqrt{e} + o(1))n(\log_2 n)^2.$$

Thus $M_3(n) = n^{1+o(1)}$ and $M_4(n) = n^{1+o(1)}$.

The upper bounds use published permutation-design constructions; they are not claims of new bounds for those designs. The lower bound is the independent Gram/path-energy argument in `../approximate/three_wise_face_lower_bound.tex`.

Lemma 2 (A triple-minimum design gives three-wise fair dice). *Let $\rho_1, \dots, \rho_q$ be permutations written as ordered lists of $[n]$. Suppose that, for every three distinct labels, each is the earliest of the three in exactly $q/3$ of these lists. Then*

$$W = P_1 \cdots P_q, \quad P_i = \rho_i \operatorname{rev}(\rho_i),$$

is three-wise fair and has $2q$ copies of each letter. The order of its q blocks is arbitrary.

Proof. Each palindrome block has two copies of each label and exactly two occurrences of either ordering of any pair. For a target triple abc , its count in P_i is zero if b is earliest in ρ_i , and two otherwise. Thus the sum of its within-block counts is $4q/3$, independent of abc . The contributions using two distinct blocks depend only on their one- and two-letter counts, and those using three distinct blocks depend only on their one-letter counts. They too are independent of the target order. Hence all six orders of any triple have the same count. The one- and two-letter statements follow in the same way. $\square$

Lemma 3 (Reversal raises odd strength by one). *If a word W is fair on every subset of at most $2s - 1$ labels, then $W \operatorname{rev}(W)$ is fair on every subset of at most $2s$ labels. Equal composition is not needed. Every multiplicity doubles.*

Proof. Work in the associative algebra in which words with repeated letters vanish, truncated above degree $2s$. Put $H = \sum_i c_i X_i$, where c_i are the letter counts. Its positive-word signature is the product of $1 + X_i = \exp X_i$ in word order. The hypothesis says

$$\log S(W) = H + E_{2s}$$

in this truncated algebra. The logarithm is a Lie element, since it is obtained by the Baker–Campbell–Hausdorff formula. Reversal acts on a homogeneous Lie element of degree j by $(-1)^{j-1}$, as follows by induction on brackets. Thus

$$\log S(\text{rev}(W)) = H - E_{2s}.$$

All commutators involving E_{2s} have degree greater than $2s$. Consequently $\log S(W \text{rev}(W)) = 2H$, precisely the signature condition for fairness through degree $2s$. $\square$

Proof of Theorem ??. A perfect sequence covering array of strength three is a multiset of q permutations in which every ordering of any triple occurs $q/6$ times. In particular it satisfies Lemma ?. Published constructions give

$$q \leq (3\sqrt{e} + o(1))n(\log_2 n)^2;$$

see Iurlano [?, Remark 5, equation (11)], using the finite- geometry construction of Tarui, Itoh, and Takei with the improved base case. Lemma ?? gives $2q$ faces for three-wise fairness, and Lemma ?? gives $4q$ for four-wise fairness. The three-wise Gram/path-energy lower bound cited above gives $M_3(n) \geq (n+3)/2$ for $n \geq 5$. $\square$

Questions exposed by the reduction. Can the logarithmic factor be removed for three-wise fair dice? Does $M_3(n) = \Theta(n)$ hold, perhaps even without a linear-size permutation design? The same question for $M_4(n)$ is equivalent up to a factor of two by Lemma ?. More generally that lemma gives $M_{2s-1}(n) \leq M_{2s}(n) \leq 2M_{2s-1}(n)$. The next distinct regime is five-wise fairness. It is natural to ask whether $M_5(n)$ is nearly quadratic, but no such lower or upper bound is asserted here. These fixed-strength questions do not settle the full-strength case $k = n$ in the manuscript.

References

- [1] E. Iurlano, *Growth of the perfect sequence covering array number*, Designs, Codes and Cryptography 91 (2023), 1487–1494, doi:10.1007/s10623-022-01168-3.
- [2] R. Yuster, *Perfect sequence covering arrays*, Designs, Codes and Cryptography 88 (2020), 585–593; author’s text.